
Study Guide

Web Programming 181

Academic Year 2025



“Research has shown that it takes 31 days of conscious effort to make or break a habit. That means, if one practices something consistently for 31 days, on the 32nd day it does become a habit. Information has been internalized into behavioral change, which is called transformation.”

Shiv Khera



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Academic Year 2025

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MODULE DESCRIPTION

Module Name	Web Programming 181
Module Code	WPR181
Qualification	BCOM
NQF Level	9
Duration (weeks)	3
Pre-requisites	None

OUTCOMES**Purpose**

The purpose of this course is to enable the student to design and develop web pages and/or websites. The emphasis will be on layout, structure, and content presentation. Good design principles will be covered, as well as the integration of multimedia elements in web pages. Additionally, the course will cover basic web server concepts and the HTTP paradigm.

Towards the end, the course introduces Content Management Systems. This helps with modular development and deployment of websites. This part builds on HTML and CSS.

Outcomes

Upon successful completion of this module, the student will be able to demonstrate:

- An informed understanding of the core area of web development including an informed understanding of its key terms, principles, and concepts such as network protocols and web servers.
- The ability to use standard mark-up, styling techniques, visual elements, and development tools to create and deploy a website on a web server.
- The ability to understand and solve a given problem to create a web-based application.
- The ability to gather information from a range of resources, including but not limited to the internet, to select and evaluate information appropriate to represent specific content on a website.
- The ability to communicate information reliably, accurately and coherently, using terms and conventions appropriate to web design and development in a practical demonstration.
- An understanding of and respect for conventions around copyright and plagiarism.

STUDENT SUPPORT

Please contact your lecturer for subject-related support. The lecturers presenting this subject are:

- Reason Sithole – sithole.r@belgiumcampus.ac.za
- Ikraam Sadek – sadek.i@belgiumcampus.ac.za
- Duan Steyn – steyn.d@belgiumcampus.ac.za
- Melary Magorimbo – Magorimbo.m@belgiumcampus.ac.za
- Asiphe Magaudini – Magaudini.a@belgiumcampus.ac.za
- Tendai Mkwaira – Mkwaira.t@belgiumcampus.ac.za
- Oliver Lulembo – lulembo.o@belgiumcampus.ac.za

If the lecturers were unable to assist, you can also contact the cluster head for this subject:

- Ms A. Mundackal – joy.a@belgiumcampus.ac.za

Further student support services are available via the counsellors:

- Lethlabile L. Selamolela – selamolela.l@belgiumcampus.ac.za
- Mathapelo Leshilo – leshilo.m@belgiumcampus.ac.za

ASSESSMENT PLAN

ASSIGNMENTS/PROJECTS

Assignment weight:	10	Assignment due date:	10-08-2025
Project2 weight:	20	Project2 due date:	18-08-2025

TESTS

Test 1 and Test 2 weight:	35	Test 2 date:	15-08-2025
Summative test weight:	35	Summative test date:	22-08-2025

STUDENT RESOURCES

Which resources will be used during this module?

PRESCRIBED MATERIAL

Textbook 1

Wolf, J. (2023) HTML and CSS: The Comprehensive Guide

Location (Library / URL / PDF)	HTML and CSS : The Comprehensive Guide (ebSCOhost.com) https://search.ebscohost.com/login.aspx?direct=true&db=nlebk&AN=3601391&site=ehost-live&authtype=sso&custid=ns266672
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RECOMMENDED READING

Location (Library / URL / PDF)

STUDENT MATERIAL

Item	Location
Content on Moodle	The relevant Moodle course
PowerPoint slides	Distributed to students via Moodle
Exercises / Activities	Dispersed throughout the course on Moodle. Some quizzes to be hosted on AssessmentQ.

TECHNOLOGY (HARDWARE OR SOFTWARE) REQUIRED

Software/Hardware	Details e.g. version to be used (either minimum or required version)
Visual Studio Code 2023	

LESSON PLAN OUTLINE

Date	Specific outcomes (SO) to be covered / Class Activity / Assessment
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Week 1 Day 1	SO1: Introduction to Web Programming SO2: Web Application Architecture
Week 1 Day 2	SO3: Intro to HTML SO4: HTML Images, Hyperlinks and Lists.
Week 1 Day 4	SO5: Working with Internal CSS SO6: External CSS -Selector conflicts and the cascade SO7: CSS Inheritance and Selector specificity
Week 1 Day 5	Class Test 1
Week 2 Day 1	SO8: CSS Grid SO9: CSS Important and Multiple Selectors
Week 2 Day 2	SO10: HTML5 New Semantics SO11: CSS Typography SO12: CSS Display and Float SO13: CSS Positioning
Week 2 Day 3	SO14: CSS Box Model SO15: CSS Main
Week 2 Day 4	SO16: HTML5 Footer
Week 3 Day 1	Class Test 2
Week 3 Day 2	SO17: HTML Tables SO18: HTML Forms
Week 3 Day 3	SO19: Form Validation Project due
Week 3 Day 4	Project Presentations
Week 3 Day 5	Summative Test

OUTCOME BREAKDOWN

Specific Outcome 1: Introduction to Web Programming

- Structure of WPR161 module
- Introduction to Web Programming
- Back-end vs. Front-end web dev
- Tools to use.
- VS Code download & installation
- Hello World website.

Specific Outcome 2: Web Application Architecture

- Web clients and servers
- Web application N-Tier architecture
- HTTPS and URL
- HTTP Content-negotiation
- HTTP Messages

Specific Outcome 3: Intro to HTML

- Defining HTML and CSS
- Appreciating the need for HTML
- HTML tags, elements, and attributes
- Introduction to HTML5
- Overview of HTML5 semantics
- HTML5 vs. core HTML family

Specific Outcome 4: HTML Hyperlinks, Images and Lists

- Adding Images
- Using Hyperlinks
- Using Lists

Specific Outcome 5: Working with Internal CSS

- Defining CSS
- Inline CSS
- Internal CSS

Specific Outcome 6: External CSS – Selector Conflicts & Cascade

- Linking to External CSS
- CSS Selector Conflicts
- Specificity in CSS
- Defining the Cascade

Specific Outcome 7: CSS Inheritance and Selector Specificity

- CSS Selector Inheritance
- CSS Selector Specificity

Specific Outcome 8: CSS Grid

- Defining CSS Grid Layout
- Grid Layout Terminology
- CSS Grid Properties
- Explicit Grid - Columns
- Explicit Grid - Rows
- Grid Gap
- Positioning with Grid Lines
- CSS Nested Grid and Grid Area

Specific Outcome 9: CSS Important and Multiple Selectors

- Important Property

- Targeting Multiple Selectors
- Descendant Selectors

Specific Outcome 10: HTML5 New Semantic

- HTML5 Semantic Elements
- Why use Semantic Elements?

Specific Outcome 11: CSS Typography

- CSS Typography
- Font-Family
- Font-text
- Font-colour
- Font-size

Specific Outcome 12: CSS Display and Float

- CSS Display
- CSS Float

Specific Outcome 13: CSS Positioning

- CSS Positioning

Specific Outcome 14: CSS Box Model

- CSS Box Model
- CSS Padding
- CSS Margin
- CSS Border
- CSS Background

Specific Outcome 15: CSS Main

- CSS Main

Specific Outcome 16: HTML5 Footer

- HTML5 Semantic elements
- Footer Styling

Specific Outcome 17: HTML Tables

- HTML Table

Specific Outcome 18: HTML Forms

- HTML Form Elements
- Input Types

Specific Outcome 19: Form Validation

- Defining Form Validation